Insurance Claim Triage · How it all fits together

One claim goes in → a trust score, band and reasons come out → a human decides → the knowledge base learns, but only through a gate.

3 LLM agents
investigator · verifier · explainer
all three run on every claim

2 LLM helpers
reranker · evaluation judge
single model calls, not agents

4 code engines
orchestrator · rules
scorer · tool server

1 data store
Neo4j: graph + BM25
+ DuckDB data layers

1 human decides
every claim
the scorer never decides

1 INTAKE
validate and route

- **Validate input**
  Pydantic: type + event
- **Orchestrator**
  plain Python router, no LLM:
  claim type → line skill;
  > 1 active policy → + Cross-policy

EXAMPLE (ILLUSTRATIVE)
Health claim, ₹1.8 lakh, filed 3 days after policy start; the person also holds a term + a motor policy

2 GATHER
context and evidence

- **Customer-360 lookup**
  all policies + claims of the person
- **Rules + graph counts**
  dates, duplicates, shared entities
- **Hybrid retriever**
  BM25 + graph hops → RRF → top 10
- **Reranker (LLM helper)**
  listwise: top 10 → top 5

3 INVESTIGATE
one agent, several checklists

Health · Term · Motor
+ Cross-policy (if > 1 policy)

- **Investigator**
  one agent, one tool loop;
  Haiku, or Sonnet with cross-policy
- **Vetted tools**
  13 read-only tools, no labels

≤ 7 tool calls (≤ 9 with cross-policy)

4 CHALLENGE
verify and push back

- **Verifier**
  ① code check: every evidence ID really came from a tool
  ② devil's advocate: argues the legitimate case, lowers confidence
- **Verified indicators**
  evidence IDs + counter-evidence

e.g. 4 indicators in → 3 survive
result: fewer false reds
model: GPT-4o-mini, a second opinion

5 SCORE & EXPLAIN
code decides the number

trust 0.24

- **Scorer · pure Python**
  noisy-OR → trust = 1 − risk
  caps: missing evidence 0.80, exact duplicate 0.29
- **Explainer**
  reasons + evidence IDs
  "turns green if X is supplied"

🔒 the LLM never sets the score

6 HUMAN DECIDES
the reviewer's call

- **Reviewer dashboard**
  4 pages: Queue · Claim · KB · Eval

Agree · Change band · Hold

reason needed to change or hold; every decision is audited

KNOWLEDGE BASE PAGE
- ☑ same hospital, many claims
- ☑ bill 3× provider median
- ☐ weak: filed soon after start

Update knowledge base

read · *read via tools* · *ticked signals saved as pending*

KNOWLEDGE STORE · one database for the graph, the BM25 index and approved signals

Neo4j

- **Customer-360 graph**
  person · policy · claim · provider
- **BM25 full-text index**
  snippets · patterns · clauses
- **Pattern cards + clauses**
  with "legitimate exceptions"
- **Past reviewer decisions**
  precedents + provider record
- **Approved signals**
  ticked → pending → 🚦 → live

feeds the store

OFFLINE · prepare the knowledge base once, before any claim runs

① **Synthetic data**
customers · policies · claims · documents; hidden labels kept apart

② **Seeded generator**
writes graph nodes + edges
no real personal data

③ **BM25 index + pattern cards**
text snippets become searchable

④ **Loaded into Neo4j**
once, before the first claim

HOW TO READ THIS
- LLM agent
- LLM helper
- deterministic code
- human
- data store
- data flow
- store read
- store write (gated)

PLATFORM UNDER EVERY STEP

- **LiteLLM router**
  Claude Haiku + Sonnet · GPT-4o-mini verifier
  each falls back to the other vendor
- **Local run traces**
  every agent step, tool call, model, token count and latency is saved
- **Evaluation harness**
  dev / held-out cases · rules-only baseline
  LLM judge: gpt-4o (different vendor)
- **Config + contracts**
  weights.yaml · thresholds.yaml
  frozen Pydantic contracts

WHAT WE MEASURE · agent evaluation is the critical criterion (✓ done on dev and held-out · ▪ still to do)

- ✓ right card in top 3 for 35 of 36 test searches
- ✓ tools · 4.8 calls per case (≤ 7)
- ✓ every flag has evidence IDs (100%)
- ✓ fraud landing in green: 0% on the original cases, dev and held-out (target ≤ 5%)
- ✓ missing evidence never green
- ▪ same band across 3 runs
- ✓ safety check blocks a bad signal
- ✗ agents below rules on easy data (92–93% vs 99%; larger model 95% on X1–X2)